# Retail Sales & Profitability Analysis

## Overview

This project analyzes retail sales data to evaluate business performance, identify profitability drivers, and uncover actionable insights. SQL was used for data analysis, Excel for preprocessing, and Power BI for building an interactive dashboard.

## Key Metrics

- **Total Sales:** Strong overall revenue performance across years
- **Total Profit:** Positive but uneven across categories
- **Profit Margin:** Impacted significantly by discounting strategies
- **Total Orders:** Consistent transaction volume

## Key Insights

1. Category Performance
   - Technology is the most profitable category
   - Furniture shows high sales but low profitability
   - Sub-categories like Tables and Bookcases contribute to losses
2. Customer Segments
   - Consumer segment contributes the highest share (~50%)
   - Corporate and Home Office segments provide stable but lower contributions
3. Regional Analysis
   - Sales are concentrated in key states such as California and New York
   - Certain regions outperform in both revenue and profitability
4. Seasonal Trends
   - Sales follow consistent monthly patterns across years
   - Peak months indicate strong seasonal demand
5. Discount impact
   - Low discounts maintain healthy average profit
   - Discounts above 20-30% significantly reduce average profit, with higher ranges leading to losses
   - Excessive discounting leads to losses

## Business Recommendations

- Optimize discount strategies to protect profit margins
- Focus on high-performing categories like Technology
- Improve pricing and cost management to loss-making products
- Align inventory and marketing strategies with seasonal demand
- Strengthen operations in high-performing regions

## Conclusion

The analysis shows that while sales performance is strong, profitability is highly influenced by discounting strategies and product mix. A data-driven approach to pricing, product focus, and regional planning can significantly enhance overall business performance.